**Amirkabir University of Technology**
**(Tehran Polytechnic)**

**Department of Electrical Engineering**

Report for the Course Linear Control Systems
Final Project

# Analysis and design of the autopilot controller for a Bicopter

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**Table of Contents**

**Introduction:**

**1. Overview:**

In recent decades, the use of Unmanned Aerial Vehicles (UAVs) has expanded, ranging from hobbies to military applications such as mapping, maintenance and surveillance operations, transportation, and search and rescue.

These types of drones, which use propellers, are attractive due to their vertical take-off and landing (VTOL) capabilities and high speed. Quadcopters (four-propeller drones) are the most widely used in these fields.

However, quadcopters face certain limitations during indoor operations. Indoor environments typically demand a high payload capacity within a compact size, alongside high endurance. For instance, a DJI quadcopter can carry up to 0.9 kg, but its wingspan is 809 mm, which exceeds the average width of doors in indoor operations. Conversely, the Crazyflie quadcopter can navigate through small gaps but has a maximum payload capacity of only 15 grams.

A proposed solution for adapting drones to confined environments is altering their configuration without compromising their power and efficiency. A high number of propellers increases both the weight and the mechanical complexity of the drone. Consequently, this has led to the design of a drone utilizing a tilt-rotor mechanism—a type of vertical take-off aircraft that lifts off the ground using one or more rotors that can rotate up to about 90 degrees. These types of drones are Bi-copters, which, in addition to being able to pass through narrow spaces and indoor environments, have lower weight and energy consumption due to having fewer actuators and a smaller overall size.

**2. Model:**

To model the bicopter, we observe the system in a Cartesian coordinate frame with three axes: $x$, $y$, and $z$. The diagram is divided into two frames: the Earth frame, which is stationary, and the bicopter body frame. The linear position ($\Gamma^E$) of the bicopter is determined by the vector coordinates between the origin of the Earth frame and the origin of the body frame with respect to the Earth frame. The angular position of the bicopter ($\Theta^E$) is also determined by the orientation of the bicopter frame relative to the Earth frame. The equations are as follows:

$$\Gamma^E = [x \quad y \quad z]$$
$$\Theta^E = [\phi \quad \theta \quad \psi]$$

The bicopter rotates around the $x$, $y$, and $z$ axes via a rotation matrix. The bicopter's equations consist of 12 states, which are divided into two categories: 6 states for its translational motion and 6 states for its rotational motion.

$$x_{1-6} = [x \quad \dot{x} \quad y \quad \dot{y} \quad z \quad \dot{z}]^T$$
$$x_{7-12} = [\phi \quad \dot{\phi} \quad \theta \quad \dot{\theta} \quad \psi \quad \dot{\psi}]^T$$

The thrust of the right rotor ($FR$) and left rotor ($FL$), generated by the blades and rotors, along with the right tilt angle $\gamma_R$ and left tilt angle $\gamma_L$, lead us to the following equations using Newton's second law:

$$\sum F_x = F_R sin\ \gamma_R + F_L sin\ \gamma_L$$
$$\sum F_y = 0$$
$$\sum F_z = F_R cos\ \gamma_R + F_L cos\ \gamma_L$$

Given the input $u$, we can obtain the total lift force and torque of the bicopter, where in the following equations $C_T$ is the blade thrust coefficient, and $\Omega_R$ and $\Omega_L$ are the rotational speeds of the right and left rotors.

$$u = [u_1 \quad u_2 \quad u_3 \quad u_4]^T$$
$$u_1 = C_T(\Omega_R^2 \cos\ \gamma_R + \Omega_L^2 \cos\ \gamma_L)$$
$$u_2 = C_T(\Omega_R^2 \cos\ \gamma_R - \Omega_L^2 \cos\ \gamma_L)$$
$$u_3 = C_T(\Omega_R^2 \sin\ \gamma_R + \Omega_L^2 \sin\ \gamma_L)$$
$$u_4 = C_T(\Omega_R^2 \sin\ \gamma_R - \Omega_L^2 \sin\ \gamma_L)$$

The rotational subsystem (roll, pitch, yaw) forms the inner loop, and the translational motion subsystem (x, y, z (altitude)) forms the outer loop. According to the Newton-Euler dynamic solution, we have:

$$\ddot{x} = -\frac{1}{m}(s\phi s\psi + c\phi s\theta c\psi)u_1 - \frac{c\theta c\psi}{m}u_3$$
$$\ddot{y} = -\frac{1}{m}(-s\phi c\psi + c\phi s\theta s\psi)u_1 + \frac{c\theta s\psi}{m}u_3$$
$$\ddot{z} = g - \frac{1}{m}(c\phi c\theta)u_1 - \frac{s\theta}{m}u_3$$
$$\ddot{\phi} = \frac{L}{I_{xx}}u_2$$
$$\ddot{\theta} = \frac{h}{I_{yy}}u_3$$
$$\ddot{\psi} = \frac{L}{I_{zz}}u_4$$

The above parameters are defined partly in the figure and partly in the table below:

| Parameter | Symbol | Value | Unit |
|---|---|---|---|
| Mass of the UAV Bicopter | $m$ | 0.725 | $kg$ |
| Gravitational acceleration | $g$ | 9.81 | $m.s^{-2}$ |
| Vertical distance between CoG and center of the rotor | $h$ | 0.042 | $m$ |
| Horizontal distance CoG and rotor center | $L$ | 0.225 | $m$ |
| Thrust coefficient | $C_T$ | 0.1222 | $-$ |
| The Moment of Inertia along x axis | $I_{xx}$ | $0.116 \times 10^{-3}$ | $kg.m^2$ |
| The Moment of Inertia along y axis | $I_{yy}$ | $0.0408 \times 10^{-3}$ | $kg.m^2$ |
| The Moment of Inertia along z axis | $I_{zz}$ | $0.105 \times 10^{-3}$ | $kg.m^2$ |

And finally, the state equations of the bicopter are simplified as follows:

$$
\begin{aligned}
\dot{x}_2 &= \ddot{x} = -\frac{1}{m}(s\phi s\psi + c\phi s\theta c\psi)u_1 - \frac{c\theta c\psi}{m}u_3 \\
\dot{x}_3 &= \dot{y} = x_4 \\
\dot{x}_4 &= \ddot{y} = -\frac{1}{m}(-s\phi c\psi + c\phi s\theta s\psi)u_1 + \frac{c\theta s\psi}{m}u_3 \\
\dot{x}_5 &= \dot{z} = x_6 \\
\dot{x}_6 &= \ddot{z} = g - \frac{1}{m}(c\phi c\theta)u_1 - \frac{s\theta}{m}u_3 \\
\dot{x}_7 &= \dot{\phi} = x_8 \\
\dot{x}_8 &= \ddot{\phi} = \frac{L}{I_{xx}}u_2 \\
\dot{x}_9 &= \dot{\theta} = x_{10} \\
\dot{x}_{10} &= \ddot{\theta} = \frac{h}{I_{yy}}u_3 \\
\dot{x}_{11} &= \dot{\psi} = x_{12} \\
\dot{x}_{12} &= \ddot{\psi} = \frac{L}{I_{zz}}u_4
\end{aligned}
$$

Now, the above equations must be linearized around their equilibrium point. According to the definition of an equilibrium point, in a dynamic system, it refers to a point where the various states of the system, such as position, velocity, etc., are such that no further changes are generated in the system, or in other words, the system remains at rest or in equilibrium. In mathematical intuition for dynamic systems, the equilibrium point is defined as a state where the time derivatives of all system variables are zero.

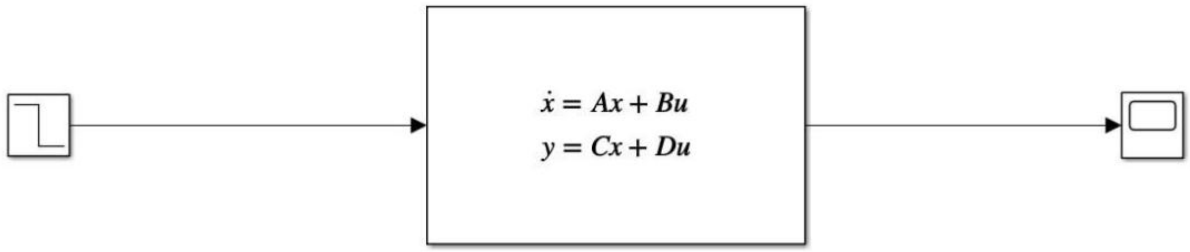
Intuitively, the equilibrium point in our system under consideration is a state where the bicopter remains hovering in a point in space, stationary and fixed, without vertical or horizontal movement. In such a state, the propellers are positioned exactly in the vertical direction, and the forces generated by the motors precisely compensate for the gravitational force of the Earth to maintain the stationary state of the bicopter.

If we want to describe it more precisely, it is necessary to mention that the angles of the propellers are all zero, the input forces in the direction of the $x$ and $y$ axes are zero, and the input force in the direction of the $z$ axis is equal to the Earth's gravitational force ($mg$).

$$u_1 \simeq mg$$
$$u_3 \simeq 0$$
$$\sin(\phi) \simeq \phi \rightarrow \phi \simeq 0, \cos(\phi) \simeq 1$$
$$\sin(\theta) \simeq \theta \rightarrow \theta \simeq 0, \cos(\theta) \simeq 1$$
$$\sin(\psi) \simeq 0 \rightarrow \cos(\psi) = \sqrt{1 - \sin^2(\psi)} = 1$$

And finally, the linear equations around the hover equilibrium point of the bicopter are simplified as follows, and after organizing, we arrive at the matrices required for the state-space representation:

$$\dot{x}_1 = \dot{x} = x_2$$
$$\dot{x}_2 = \ddot{x} = -g\theta$$
$$\dot{x}_3 = \dot{y} = x_4$$
$$\dot{x}_4 = \ddot{y} = g\phi$$
$$\dot{x}_5 = \dot{z} = x_6$$
$$\dot{x}_6 = \ddot{z} = g - \frac{u_1}{m}$$
$$\dot{x}_7 = \dot{\phi} = x_8$$
$$\dot{x}_8 = \ddot{\phi} = \frac{L}{I_{xx}} u_2$$
$$\dot{x}_9 = \dot{\theta} = x_{10}$$
$$\dot{x}_{10} = \ddot{\theta} = \frac{h}{I_{yy}} u_3$$
$$\dot{x}_{11} = \dot{\psi} = x_{12}$$
$$\dot{x}_{12} = \ddot{\psi} = \frac{L}{I_{zz}} u_4$$




$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -g & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & g & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{1}{m} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \frac{L}{I_{xx}} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{h}{I_{yy}} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{L}{I_{zz}} \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

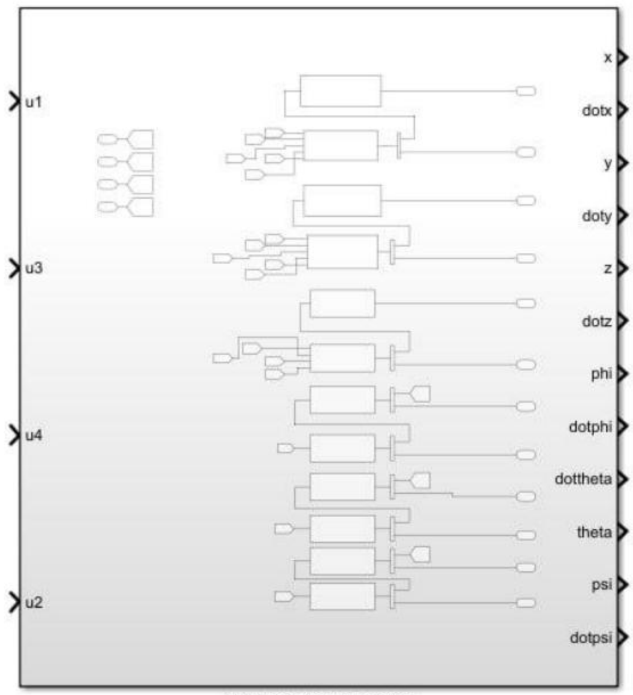
As a result, the transfer function of the linear equations is equal to:

Vertical axis from top to bottom respectively: $x, x', y, y', z, z', \Phi, \Phi', \Theta, \Theta', \Psi, \Psi'$

Horizontal axis from left to right respectively: $u_1, u_2, u_3, u_4$

Out[689]=

$$\begin{pmatrix} 0. & 0. & -\frac{10098.5}{s^4} & 0. \\ 0. & 0. & -\frac{10098.5}{s^3} & 0. \\ 0. & \frac{19028.}{s^4} & 0. & 0. \\ 0. & \frac{19028.}{s^3} & 0. & 0. \\ \frac{1.37931}{s^2} & 0. & 0. & 0. \\ \frac{1.37931}{s} & 0. & 0. & 0. \\ 0. & \frac{1939.66}{s^2} & 0. & 0. \\ 0. & \frac{1939.66}{s} & 0. & 0. \\ 0. & 0. & \frac{1029.41}{s^2} & 0. \\ 0. & 0. & \frac{1029.41}{s} & 0. \\ 0. & 0. & 0. & \frac{2142.86}{s^2} \\ 0. & 0. & 0. & \frac{2142.86}{s} \end{pmatrix}$$



UAV Bicopter Dynamics

**Determining Control Objectives:**

In a linear control system, control objectives are identified in such a way that the system state reaches a desired state or the deviation from this desired state is minimized. To identify control objectives and achieve them, you can proceed as follows:

**1. System Modeling:** As we did in the previous section, we must first create a mathematical model of the system, which also includes linear differential equations.

**2. Set points:** These are the objectives that we want the system to achieve or maintain, and they act as the system reference.

**3. Controller Design:** By selecting an appropriate controller, we guide the system toward the set objectives.

**4. Simulation and Analysis:** Using simulation software, analyze the system performance to ensure that the system reaches its objective.

**5. Parameter Tuning:** Tune the controller parameters to guarantee the optimal performance of the system.

Control objectives for the bicopter are to have a stable and accurate flight:

**1. Stability:** Ensuring the stability of the docopter during flight by controlling its pitch, roll, and yaw, which is very important to prevent tipping over or uncontrollable rotation of the bicopter. In control systems, we can achieve this goal by ensuring that all poles of the transfer function are on the left side.

**2. Altitude Control:** Maintaining the desired altitude by adjusting the force generated by the motors. This includes controlling the vertical position and ensuring a smooth take-off and landing.

**3. Position Control:** Achieving accurate horizontal movements and maintaining the desired position in space. This includes controlling the forward, backward, and lateral movements of the bicopter.

**4. Rotation Control:** Adjusting the orientation of the bicopter in terms of pitch, roll, and yaw angles. Attitude control ensures that the bicopter follows the desired angular trajectories and remains level during flight.

**5. Speed Control:** Managing the speed of the bicopter in different directions. This includes linear speed control for smooth and controlled movements.

**6. Disturbance Rejection:** Minimizing the impact of external disturbances such as wind gusts or sudden load changes. The control system must compensate for these disturbances to maintain stable flight.

**7. Hovering and Maintaining a Constant Position in the Air:** This includes precise control of all axes to keep the bicopter at a fixed point.

These control objectives are achieved through a combination of sensors, control algorithms, and feedback loops that continuously monitor the status of the bicopter and adjust the motor inputs accordingly. In this paper, we address sections 2 and 7; for the other sections, non-linear control is used due to the complexity.

**PID Compensator:**

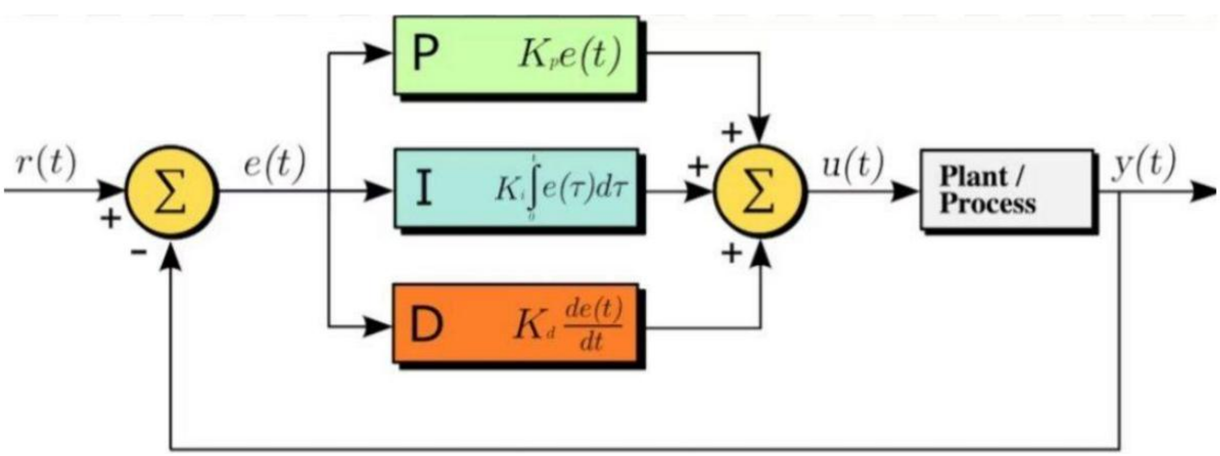
Among the reasons we need a compensator is that, according to the current transfer function of the bicopter, the system is normally unstable due to the presence of poles at the origin, and we require feedback. To achieve the desired characteristics for settling time, overshoot, etc., we use a PID compensator. $U(t)$ is the controller output, $e(t)$ is the error (the difference between the desired value and the actual value), and $K_p$, $K_i$, and $K_d$ are the proportional, integral, and derivative coefficients, respectively.

Laplace Domain:

$$C(s) = k_P + k_D s + k_I \frac{1}{s}$$

Time Domain:

$$u(t) = k_P e(t) + k_D \frac{de(t)}{dt} + k_I \int_0^t e(\tau) d\tau$$




A PID (Proportional-Integral-Derivative) controller introduces a pole at the origin, which increases the system type and is suitable for eliminating steady-state error. It consists of three components, each playing a specific role in tuning the system's response:

P (Proportional): This part operates based on the current system error. That is, the controller output is directly proportional to the existing error, and increasing $K_p$ can reduce the rise time.

I (Integral): This part acts based on the accumulation of past errors. That is, the controller output includes the integral of the errors over time, and increasing $K_i$ helps reduce or eliminate the steady-state error.

D (Derivative): This part operates based on the rate of change of the error. That is, the controller output includes the derivative of the error with respect to time, which helps control future error. Increasing $K_d$ can reduce oscillations and the percentage overshoot, thereby improving system stability.

Compensator Design:

First, we extract the required transfer functions using the table from the previous pages. By taking into account the desired target parameters for our bicopter, we calculate and obtain the gain values for our PID controller. Following this, the simulation is performed and executed alongside the corresponding MATLAB codes.

The primary performance specifications are selected to enforce a maximum overshoot ($\%OS$) of 5% and a settling time ($T_s$) of 1 second. Evaluating the decoupled open-loop transfer functions for altitude and attitude loops yields:

$$\frac{Z(s)}{u_1(s)} = \frac{1.37931}{s^2}$$

$$\frac{\Phi(s)}{u_2(s)} = \frac{1939.66}{s^2}$$

$$\frac{\Theta(s)}{u_3(s)} = \frac{1029.41}{s^2}$$

$$\frac{\Psi(s)}{u_4(s)} = \frac{2142.86}{s^2}$$

Using standard second-order system design equations, the target damping ratio ($\zeta$) and natural frequency ($\omega_n$) are calculated:

$$\%OS = e^{-\left(\frac{\zeta\pi}{\sqrt{1-\zeta^2}}\right)} \times 100 \implies \zeta \simeq 0.69$$

$$T_s = \frac{4}{\zeta\omega_n} = 1.0 \implies \omega_n \simeq 5.797 \text{ rad/s}$$

This places the dominant complex conjugate closed-loop poles at a real value of approximately $-4$. Because the open-loop plant features a double integrator at the origin ($s^2$), a closed-loop system combined with a PID controller results in a Type-3 tracking loop.

To maintain dominant second-order behavior, a third far-left pole is placed at $\alpha = 40$ (ten times farther into the left half-plane) to minimize its effect on transient response. The desired characteristic equation is given by:

$$(s^2 + 2\zeta\omega_n s + \omega_n^2)(s + \alpha) = (s^2 + 8.12s + 33.64)(s + 40)$$

Analytically resolving this equation yields the initial target PID gains. These parameters were implemented and simulated using MATLAB. Due to the unmodeled high-order cross-coupling present in numerical tests, the analytical gains exhibited slight performance deviations. Consequently, optimization loops and heuristic tuning were executed to refine the final gains and meet the strict transient criteria.

**Analysis of Disturbances and Noise**

In control systems, disturbances and noise are two types of unwanted inputs that can affect the performance and accuracy of the system, while both cause errors.

**Noise**

Origin: High-frequency electrical fluctuations produced by onboard sensors, measurement instruments, or internal electronic circuitry.

Characteristics: Random high-frequency variations that obscure the true state

values of the measured outputs.
Impact: Directly degrades feedback signal integrity, feeding corrupt data back to the controller and leading to inefficient actuator activity.

**Disturbance**
Origin: Environmental factors that act directly on the physical structure of the drone.
Characteristics: Systematic or random low-frequency variations that displace the physical position or orientation of the vehicle.
Impact: Displaces the physical drone from its trajectory. The controller must actively exert counter-forces to preserve tracking stability.

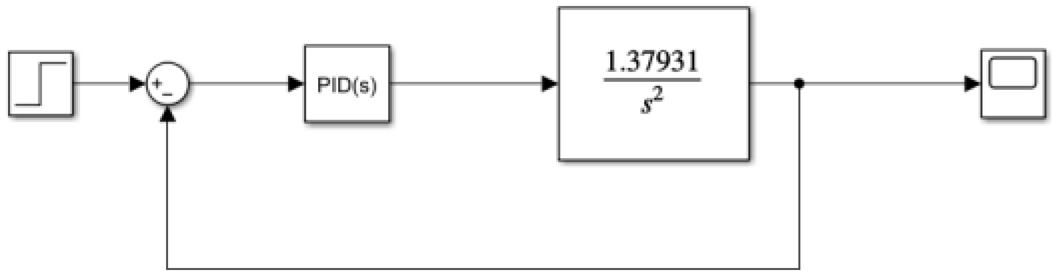
While high-frequency sensor noise is typically mitigated using low-pass filtering topologies, low-frequency external disturbances are managed via robust controller synthesis and integral action.

**Simulation Profiles**
To evaluate the bi-copter's hovering performance, three distinct operational scenarios were simulated within MATLAB/Simulink:

- Altitude Track Regulation: Assessing how effectively the vehicle transitions from a stationary ground state to a targeted hover altitude.
- External Disturbance Rejection: Inducing arbitrary impulse changes in pitch ($\theta$) and roll ($\phi$) angles to measure how rapidly the system restores equilibrium.
- Sensor Noise Attenuation: Measuring system robustness against high-frequency measurement noise.

For altitude control, a full PID topology was chosen over a standard PD design because its integral action is essential for eliminating steady-state tracking errors caused by continuous external disturbances.

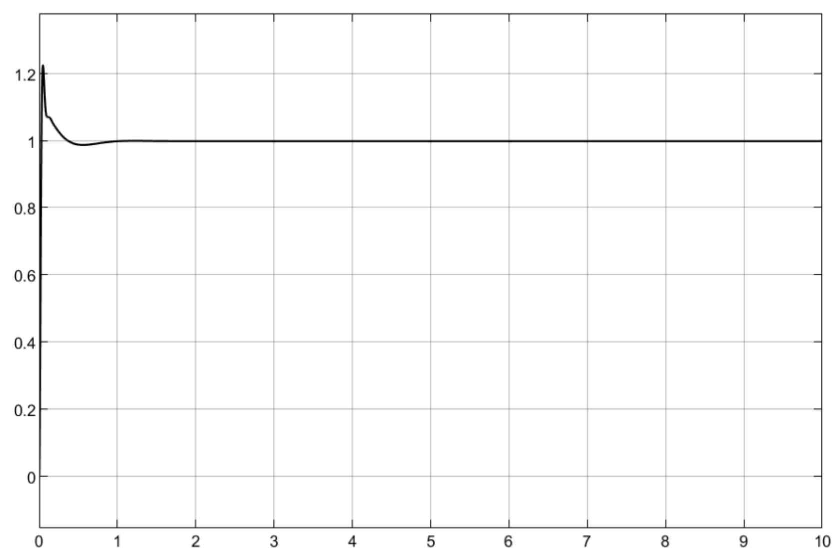
The corresponding block diagram layout developed within Simulink is structured as follows:




Because the system transfer function's denominator is third-order, assuming a far-off pole at -40 provides a clear analytical solution for the PID controller gains to fulfill the required design parameters.

| KP | KI | KD |
|---|---|---|
| 364 | 1777.776 | 48 |

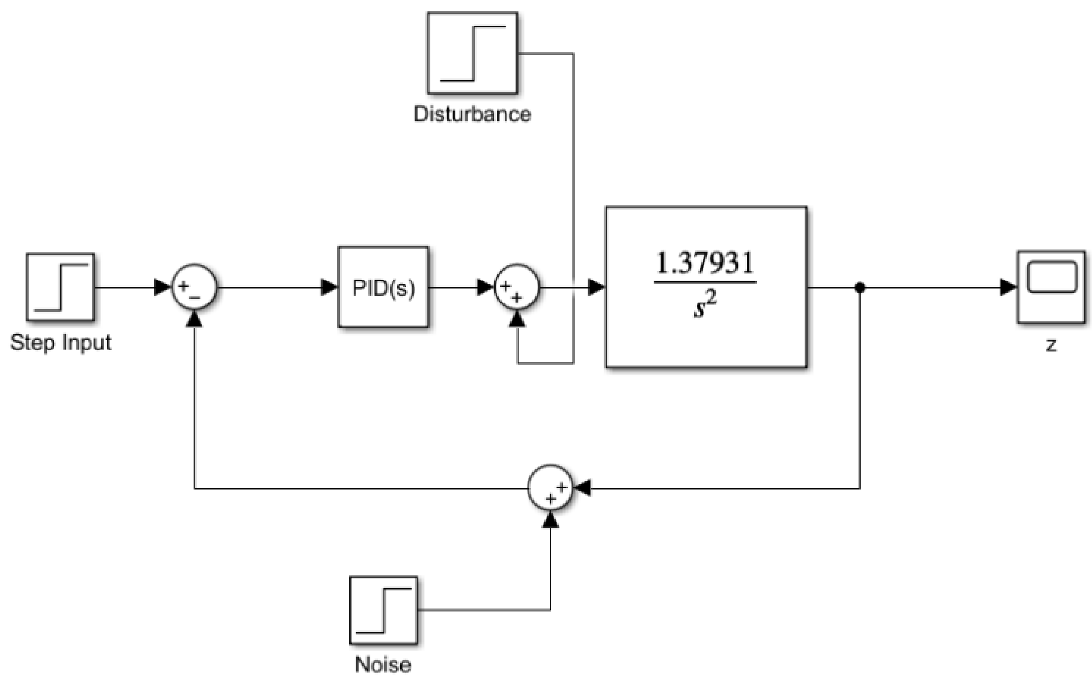
And step response is like:



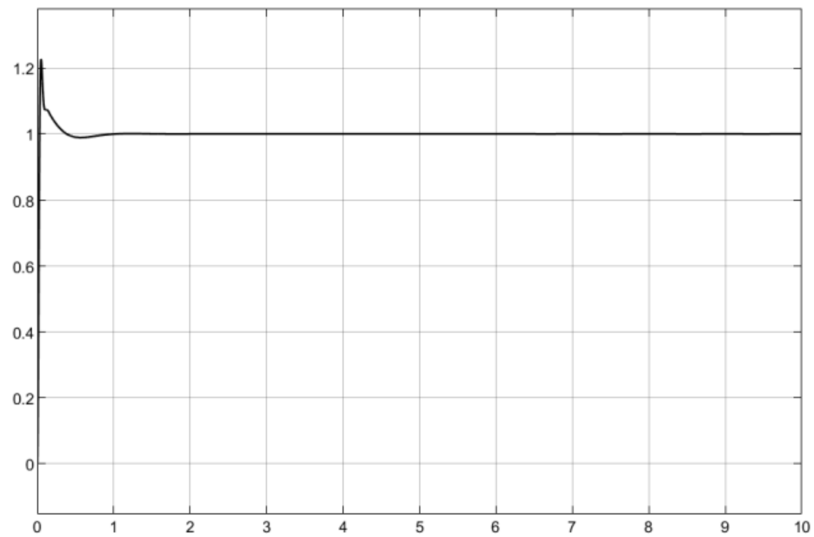
As we can see, we have a **20% overshoot** and a **settling time** of less than one second, which is desirable for us.

**The effect of disturbance and noise on altitude:**

Disturbance and noise enter the system as follows:

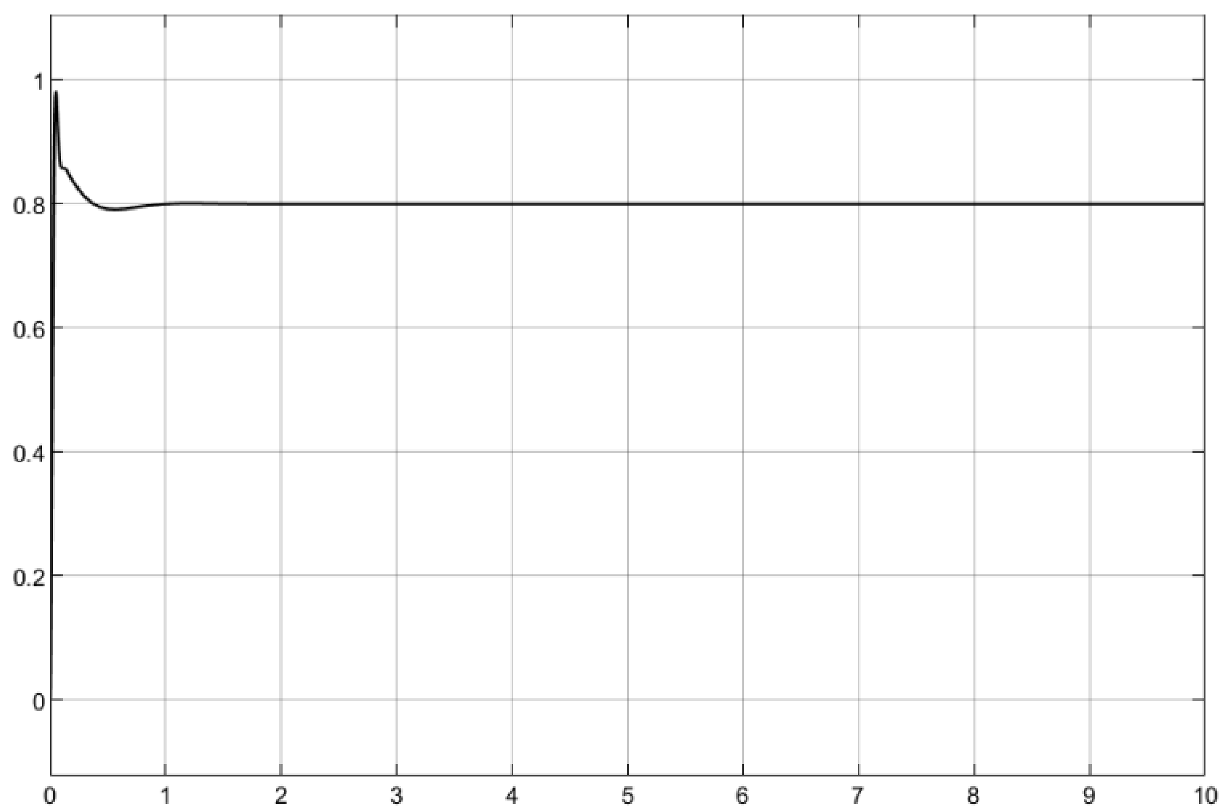



In the first stage, we analyze the system's response to a **step input** under conditions where a **step disturbance** is present.




We see that no significant effect was produced, which is one of the advantages of the PID controller. If a PD controller were used, adding the disturbance would introduce a steady-state error.

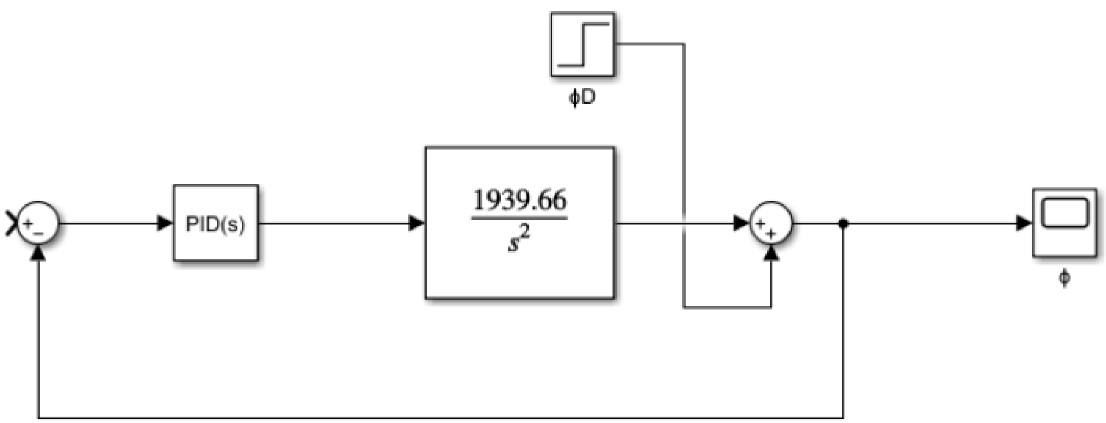
Now, we see the response to a **step input** and **step noise** (with a magnitude of 0.2).



As is evident, the noise has a direct impact on the output, and whatever input we provided to the noise, we observed the exact same behavior at the output.

To prevent the effect of noise in the real world, a filter can be used to eliminate the noise, which exhibits high-frequency characteristics.

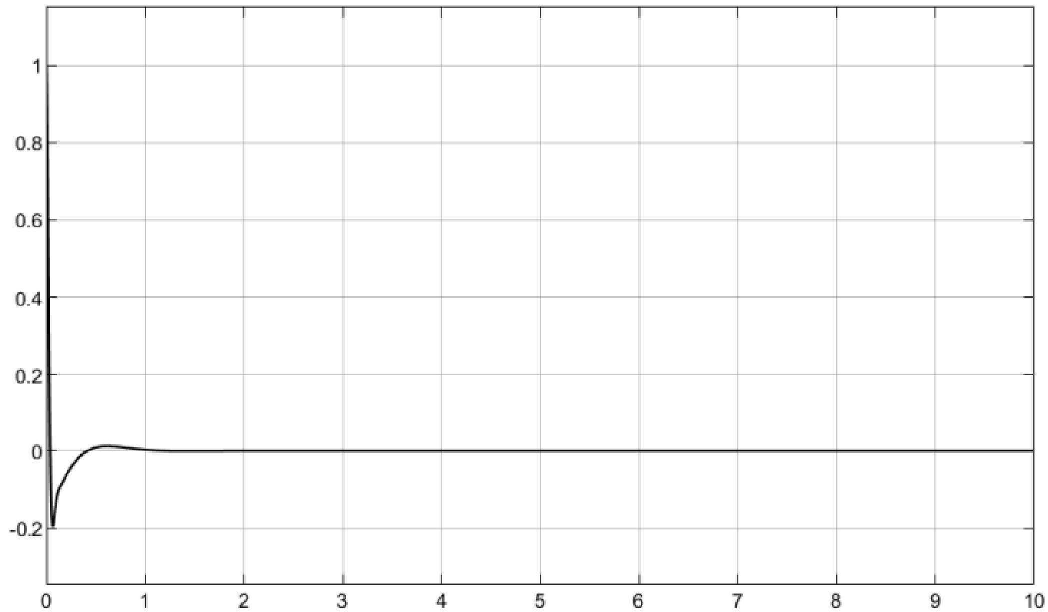
Angle (Attitude) Control: Since our objective in angle control is to keep the robot steady in a state where $\theta$ and $\phi$ are equal to zero, we assume the system as follows:




In which we must compensate for the effect of $\phi D$ on the output. After obtaining the transfer function, we calculate the PID gains.

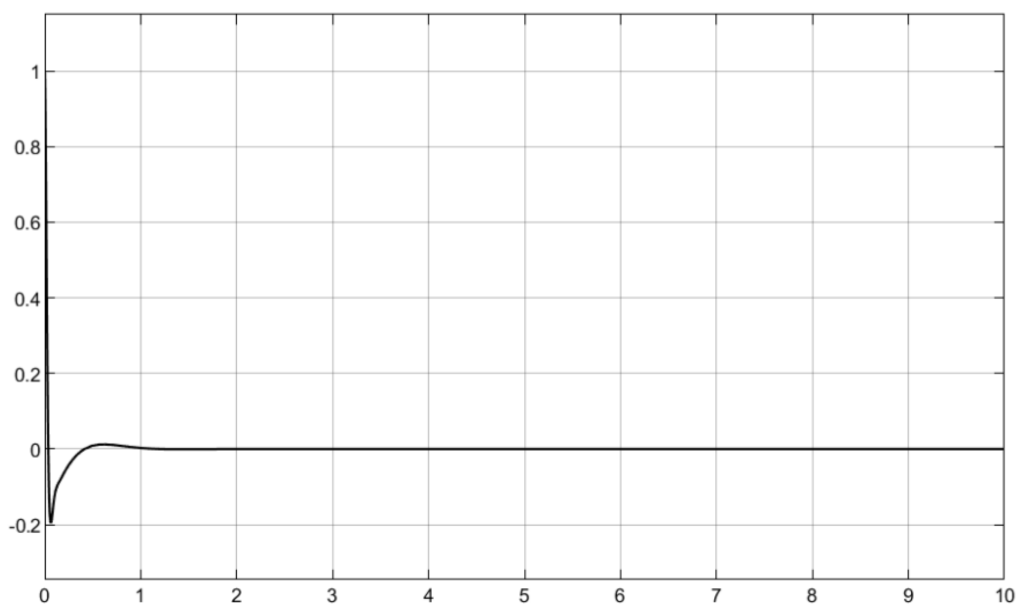
| KP | KI | KD |
|---|---|---|
| 0.182 | 0.693 | 0.0247 |

And with these values, the output plot becomes as follows:

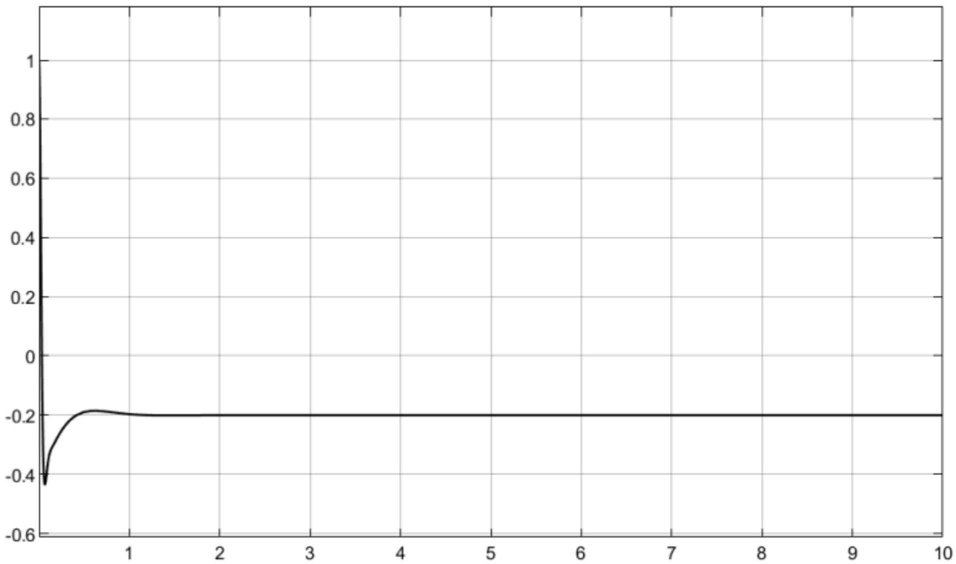



As we can see, the **angle** reaches the **desired value** in less than one second, with a **20% overshoot**.

Plot after applying the **disturbance** before the **Plant**:




Plot after applying **noise**:




We see that the effect of noise and disturbance on the angle is exactly the same as its effect on the altitude. To control the angle $\theta$, we can proceed as before; the PID coefficients [gains] simply scale by the same ratio due to the similarity of the transfer functions. Given that the transfer functions of both systems are identical, the impact of noise and disturbance on both is the same.

| KP | KI | KD |
|---|---|---|
| 0.3435 | 1.3058 | 0.0466 |

To control $\psi$ [yaw], since we have full control over it, we can control it just like $z$ [altitude].

| KP | KI | KD |
|---|---|---|
| 0.234 | 1.1443 | 0.0308 |

To control the **non-linear system**, if we use a **linear controller**, the system becomes highly **unstable**.

**References:**


1. Attitude Control and Low Cost Design of UAV Bicopter FAHMIZAL1,°,◊, HANUNG ADI NUGROHO2,° , ADHA IMAM CAHYADI3,° , and IGI ARDIYANTO4,° °Department of Electrical and Information Engineering, Engineering Faculty, Universitas Gadjah Mada, 55281, Indonesia ◊Department of Electrical Engineering and Informatics, Vocational College, Universitas Gadjah Mada, 55281, Indonesia
2. Trajectory Tracking Control of UAV Bicopter using Linear Quadratic Gaussian FAHMIZAL1, °,◊, HANUNG ADI NUGROHO2,° , ADHA IMAM CAHYADI3,° , and IGI ARDIYANTO4,° °Department of Electrical and Information Engineering, Engineering Faculty, Universitas Gadjah Mada, 55281, Indonesia ◊Department of Electrical Engineering and Informatics, Vocational College, Universitas Gadjah Mada, 55281, Indonesia
3. Performance Evaluation of Balancing Bicopter using P, PI, and PID Controller Esa Apriaskar1, Fahmizal2, Nur Azis Salim3 , and Dhidik Prastiyanto4 1,3,4Department of Electrical Engineering, Universitas Negeri Semarang Gunungpati, Semarang, 50229, Indonesia 2Department of Electrical Engineering and Informatics, Vocational College, Universitas Gadjah Mada Jl. Yacaranda, Sekip Unit IV, Yogyakarta, 55281, Indonesia
4. https://youtu.be/UR0hOmjaHp0?si=qeHogSAG9BdbtoQt